

Year 7 · Chapter 2 · Angles: answers (teacher copy)

2.01 Naming angles (booklet p. 3)

WE DO examples

- Ex 1: $\angle ABC$ or $\angle CBA$
- Ex 2: vertex K in the middle
- Ex 3: $\angle QOR$
- Ex 4: $\angle QOR$, $\angle ROP$, $\angle QOP$
- Ex 5: $\angle LON$
- Ex 6: There are several angles at B, so we need 3 letters

Set A

- $\angle PQR$
- $\angle DEF$
- $\angle XYZ$
- $\angle MNK$
- $\angle STU$
- $\angle GHJ$

Set B

- vertex P, arms PF and PR
- vertex Z, arms ZM and ZQ

Set C

- $\angle DBE$
- $\angle DBE$
- $\angle CBA$
- 6
- $\angle DBA$
- $\angle CBE$

Set D

- $\angle LMN$
- $\angle QRS$
- Yes: the vertex B is in the middle of both
- e.g. two angles sharing arm QR

Extension

- 10
- $\angle AOB$, $\angle AOC$, $\angle AOD$, $\angle BOC$, $\angle BOD$, $\angle COD$

Exit ticket

- Level 1: $\angle TUV$
- Level 2: U, because it is the point where the arms meet
- Level 3: two angles sharing arm BC

2.02 Measuring and drawing angles

(booklet p. 7)

WE DO examples

- Ex 1: 64°
- Ex 2: accurate 84° angle
- Ex 3: 128°
- Ex 4: accurate 117° angle
- Ex 5: draw 150° on the other side
- Ex 6: draw 60° on the other side

Set A

- 35°
- 72°
- 90°
- 105°
- 150°
- 48°

Set B

- 41° drawn
- 150° drawn

Set C

- 96° drawn
- 23° drawn
- 165° drawn
- 72° drawn

Set D

- $360 - 195 = 165^\circ$ on the other side
- three-quarter turn
- 230°
- 215°

Extension

- 150° and 90°
- 75°

Exit ticket

- Level 1: 58°
- Level 2: accurate 135°
- Level 3: draw 110° the other way

2.03 Classifying angles (booklet p. 11)

WE DO examples

- Ex 1: acute, right, obtuse, straight, reflex, revolution
- Ex 2: $\angle WIH$, obtuse
- Ex 3: obtuse
- Ex 4: 270° , reflex
- Ex 5: $\angle DRA$ acute (70°), $\angle ARB$ obtuse (110°)
- Ex 6: No: $90^\circ + \text{more than } 90^\circ$ is already more than 180°

Set A

- acute
- right
- obtuse
- straight
- reflex
- acute

Set B

- acute
- obtuse
- reflex
- revolution
- obtuse
- reflex

Set C

- $\angle GLU$ acute
- $\angle VRP$ obtuse
- $\angle PNE$ right
- $\angle SMA$ reflex
- $\angle ZXY$ straight
- $\angle MVT$ acute

Set D

- 120° , obtuse
- Yes, e.g. $60^\circ + 50^\circ = 110^\circ$
- 100° , 125° , 150° , 175°
- 19 (201° to 219°)

Extension

1. 900°
2. 3 revolutions

Exit ticket

1. Level 1: acute
2. Level 2: reflex; straight
3. Level 3: $\angle DRA$ acute; $\angle ARB$ obtuse

2.04 Complementary and supplementary angles (booklet p. 15)

WE DO examples

1. Ex 1: 55° ; 12°
2. Ex 2: 135° ; 80°
3. Ex 3: $m = 62$
4. Ex 4: $k = 133$
5. Ex 5: $x = 29$
6. Ex 6: 90° each

Set A

1. 55°
2. 12°
3. 86°
4. 30°
5. 45°
6. 1°

Set B

1. 135°
2. 80°
3. 2°
4. 90°
5. 168°
6. 45°

Set C

1. $m = 62$
2. $k = 133$
3. $a = 29$
4. $y = 58$
5. $p = 15$
6. $b = 90$

Set D

1. 55° and 35°
2. 36°
3. $x = 40$
4. No: it is already more than 90°

Extension

1. 60°
2. 40°

Exit ticket

1. Level 1: 28°
2. Level 2: 62°
3. Level 3: $x = 45$

2.05 Angles at a point and vertically opposite angles (booklet p. 19)

WE DO examples

1. Ex 1: $a = 110$
2. Ex 2: $m = 25$
3. Ex 3: $a = 130$
4. Ex 4: $x = 120$
5. Ex 5: $p = 145$
6. Ex 6: $p = 155$, $q = 25$, $r = 155$

Set A

1. $y = 145$
2. $a = 80$
3. $m = 44$
4. $x = 98$
5. $k = 25$
6. $d = 90$

Set B

1. $y = 35$
2. $m = 80$
3. $f = 60$

Set C

1. $f = 110$
2. $t = 83$
3. $x = 90$

Set D

1. $x = 132$ (straight line), $y = 48$ (vertically opposite)
2. $t = 173$ (angles at a point)
3. 120° each
4. 115° , 65° , 115°

Extension

1. 30° , 60° , 90°
2. $x = 30$

Exit ticket

1. Level 1: 116
2. Level 2: 38°
3. Level 3: 135

2.06 Parallel and perpendicular lines

(booklet p. 23)

WE DO examples

1. Ex 1: $AB \parallel DC$ (or $AD \parallel BC$); $AB \perp BC$
2. Ex 2: AB is perpendicular to CD ; EF is parallel to GH
3. Ex 3: construction
4. Ex 4: construction
5. Ex 5: construction
6. Ex 6: e.g. shelves ($\parallel$), wall and floor ($\perp$)

Set A

1. DC
2. BC or AD
3. AD
4. No (not in a rectangle unless it is a square)
5. at B
6. $AD \parallel BC$

Set B

1. True
2. True
3. False
4. True

Set C

1. construction
2. construction

Set D

1. construction
2. square construction

Extension

1. construction
2. every point on it is the same distance from A and B

Exit ticket

1. Level 1: $\parallel$
2. Level 2: e.g. wall and floor
3. Level 3: construction

2.07 Corresponding angles (booklet p. 27)

WE DO examples

1. Ex 1: bottom line, upper right
2. Ex 2: $a = 65$
3. Ex 3: $m = 110$
4. Ex 4: the angles sit in an F
5. Ex 5: $x = 125$, $y = 55$
6. Ex 6: Only when the lines are parallel

Set A

1. bottom ul
2. top lr
3. bottom ll
4. top ur
5. bottom lr
6. top ul

Set B

1. $a = 65$
2. $k = 50$
3. $d = 110$

Set C

1. $b = 35$
2. $m = 115$
3. $q = 118$ (corresponding 62° , then straight line)
4. $x = 122$
5. $y = 75$ (vertically opposite, then corresponding)
6. $t = 140$

Set D

1. a line crossing two or more lines
2. drawing
3. No
4. angles on a straight line

Extension

1. $y = 40$
2. $a = 30$

Exit ticket

1. Level 1: 60
2. Level 2: an F
3. Level 3: $x = 20$

2.08 Alternate angles (booklet p. 31)

WE DO examples

1. Ex 1: bottom line, upper right
2. Ex 2: $x = 60$
3. Ex 3: $k = 115$
4. Ex 4: True
5. Ex 5: $a = 48$, $b = 132$
6. Ex 6: the angles sit inside a Z

Set A

1. bottom ul
2. top ll
3. bottom ur
4. top lr
5. bottom ul
6. top ll

Set B

1. $x = 70$
2. $p = 125$
3. $w = 38$

Set C

1. $c = 65$
2. $t = 112$
3. $d = 130$ (alternate 50° , straight line)
4. $e = 73$ (corresponding)
5. $g = 140$
6. $h = 100$ (alternate 80° , then straight line)

Set D

1. drawing
2. $x = 45$
3. alternate angles and angles on a straight line
4. It is perpendicular to both lines ($y = 45$)

Extension

1. $a = 25$
2. $m = 35$; both 105°

Exit ticket

1. Level 1: 65
2. Level 2: Z
3. Level 3: $x = 35$

2.09 Co-interior angles (booklet p. 35)

WE DO examples

1. Ex 1: bottom line, upper right
2. Ex 2: $y = 60$
3. Ex 3: $m = 105$
4. Ex 4: No: they add to 180° (unless both are 90°)
5. Ex 5: $p = 65$, $q = 115$
6. Ex 6: corresponding (equal), alternate (equal), co-interior (add to 180°)

Set A

1. bottom ul
2. top lr
3. bottom ur
4. top ll
5. top lr
6. bottom ul

Set B

1. $a = 60$
2. $x = 110$
3. $b = 145$

Set C

1. $q = 62$
2. $k = 80$
3. $z = 123$ (co-interior 57° , then straight line)
4. $m = 64$ (corresponding)
5. $x = 130$
6. $y = 110$ (corresponding)

Set D

1. $x = 60$
2. 90° each; it is perpendicular to the parallel lines
3. drawing
4. $x = 60$

Extension

1. 80° and 100°
2. $a = 50$

Exit ticket

1. Level 1: 70
2. Level 2: 180°
3. Level 3: $y = 20$

2.10 Finding unknown angles on parallel lines (booklet p. 39)

WE DO examples

1. Ex 1: $a = 65$ (corresponding then straight line; or co-interior with 115° pair)
2. Ex 2: $m = 35$ (alternate)
3. Ex 3: $x = 37$ (corr.), $y = 143$ (straight line)
4. Ex 4: $k = 65$ (alternate)
5. Ex 5: $p = 62$, $m = 62$, $z = 118$
6. Ex 6: use two rules in steps, e.g. vertically opposite then alternate

Set A

1. $a = 65$
2. $b = 35$
3. $k = 65$
4. $d = 125$
5. $x = 50$
6. $q = 118$

Set B

1. $y = 132$ (alternate, then straight line)
2. $t = 110$
3. $x = 123$

Set C

1. $p = 62$, $m = 62$, $z = 118$
2. $t = 68$, $d = 112$, $a = 68$
3. $x = 130$, $y = 50$

Set D

1. four of 57° and four of 123°
2. 135° and 45°
3. The lines are parallel so each angle has a matching corresponding angle
4. any correct question

Extension

1. $x = 16$
2. $x = 40$

Exit ticket

1. Level 1: 58
2. Level 2: alternate (vertically opposite then corresponding)
3. Level 3: four 58° and four 122°

2.11 Proving lines are parallel (booklet p. 43)

WE DO examples

1. Ex 1: Yes: co-interior angles $135 + 45 = 180$
2. Ex 2: No: $110 + 72 = 182$
3. Ex 3: Yes: corresponding angles equal
4. Ex 4: $x = 95$
5. Ex 5: corresponding equal, alternate equal, co-interior supplementary
6. Ex 6: we use the angle to prove the lines are parallel (the reverse of the rule)

Set A

1. yes
2. no
3. yes
4. no
5. yes
6. no

Set B

1. Yes (corresponding equal)
2. No (alternate $50 \neq 52$)
3. Yes (co-interior $115 + 65 = 180$)

Set C

1. 112
2. 47
3. 85
4. 55
5. 70
6. 37

Set D

1. alternate angles equal, so parallel
2. $108 + 72 = 180$, co-interior supplementary, so parallel
3. No: corresponding angles are not equal
4. e.g. corresponding angles 60° and 70°

Extension

1. $x = 132$; $132 + 48 = 180$, so yes (co-interior)
2. Measurements are never exact: a proof uses the angle rules

Exit ticket

1. Level 1: Yes
2. Level 2: No: $105 + 85 = 190$, not 180
3. Level 3: $x = 37$